

Ultrasound Molecular Imaging and Its Applications in Cancer Diagnosis and Therapy

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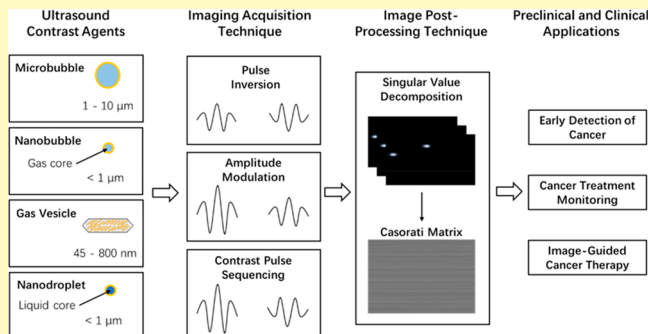
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ABSTRACT: Ultrasound imaging is regarded as a highly sensitive imaging modality used in routine clinical examinations. Over the last several decades, ultrasound contrast agents have been widely applied in ultrasound molecular cancer imaging to improve the detection, characterization, and quantification of tumors. To date, a few new potential preclinical and clinical applications regarding ultrasound molecular cancer imaging are being investigated. This review presents an overview of the various kinds of ultrasound contrast agents employed in ultrasound molecular imaging and advanced imaging techniques using these contrast agents. Additionally, we discuss the recent enormous development of ultrasound contrast agents in the relevant preclinical and clinical applications, highlight the recent challenges which need to be overcome to accelerate the clinical translation, and discuss the future perspective of ultrasound molecular cancer imaging using various contrast agents. As a highly promising and valuable tumor-specific imaging technique, it is believed that ultrasound molecular imaging will pave an accurate and efficient way for cancer diagnosis.

KEYWORDS: molecular imaging, ultrasound imaging, contrast agents, cancer diagnosis, cancer therapy, microbubble, nanodroplet, nanobubble, gas vesicle



Ultrasound imaging is a routinely used clinical imaging modality for diagnosing and characterizing various diseases in the human body.^{1,2} Its advantages include noninvasiveness, low cost, real-time imaging feedback, deep tissue penetration, and lack of radioactive exposure to patients.³ During ultrasound imaging, the ultrasound transducer transmits an acoustic wave to tissues and receives the acoustic wave reflected by the tissues. After receiving the acoustic wave from the transducer, the wave information is calculated to generate an ultrasound image. However, the imaging penetration depth decreases as the ultrasound frequency becomes higher.^{4,5} Because of this, the harmonic imaging technique was introduced, which consists of higher harmonic signals.

Ultrasound molecular imaging enables clinicians to observe organ functions in patients and to quantify morphological and functional changes.⁶ Additionally, it enables one to personalize patient care according to the results. In terms of cancer diagnostics, ultrasound molecular imaging can identify disease and determine the location of a tumor, as the molecular contrast agents can adhere to molecules expressed on angiogenic vascular endothelial cells.⁷ As a useful imaging technique for the evaluation of cancer diagnosis and therapy, ultrasound molecular imaging can assess the severity of disease and the disease progression. Furthermore, it can help clinicians

choose the most efficient therapeutic approach based on the biological properties of the certain lesion.⁸

Recently, various types of ultrasound contrast agents have been developed for better disease diagnosis in ultrasound molecular cancer imaging.^{9,10} Microbubble contrast agents are the most clinically used ultrasound contrast agents which are widely used in routine clinical contrast-enhanced ultrasound (CEUS) imaging examinations.¹¹ The diameter of the microbubble is typically between 1 and 10 μm , and therefore, it is suitable as a contrast agent within the blood pool. However, microbubbles cannot extravasate outside the vasculature.¹² Because of this, nanobubble, gas vesicle, and nanodroplet contrast agents have been developed for imaging extravascular structures of tumors.¹³ Moreover, ultrasound contrast agents can be incorporated with various targeting ligands to detect and quantify specific disease markers.⁷ A number of binding ligands have been investigated in previous

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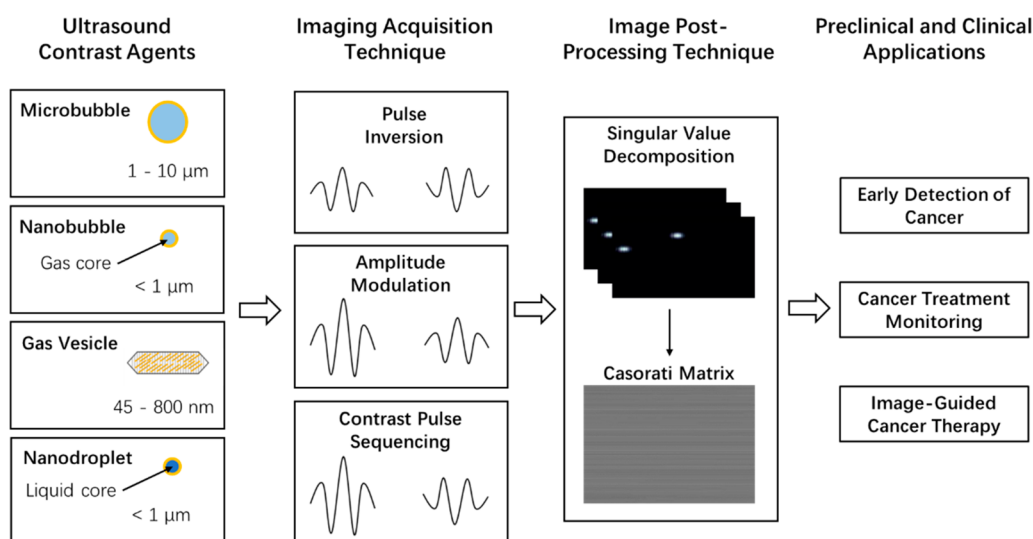


Figure 1. Overview of ultrasound contrast agents and the corresponding image acquisition and postprocessing techniques used in various preclinical and clinical applications.

Table 1. Comparison between Various Ultrasound Contrast Agents Used in Ultrasound Molecular Imaging

type	core	shell	size range	circulation lifetime	ref
microbubble	perfluorocarbon gas	phospholipid/proteins/PLGA	1–10 μm	3–5 min in humans	31
nanobubble	perfluorocarbon gas	lipid	<1 μm	3–15 min in mice	60
gas vesicle	air	proteins	45–800 nm	20–40 min in mice	29
nanodroplet	perfluorocarbon liquid	phospholipid/proteins/PLGA	<1 μm	10–60 min in mice	48

studies.^{9,14} Because of the difference in the contrast signals between ultrasound contrast agents and tissue, contrast agents can be clearly observed. Furthermore, the application of using ultrasound contrast agents along with advanced imaging acquisition techniques, for example, pulse inversion, amplitude modulation, contrast pulse sequencing, etc., has been developed to further enhance the contrast signals from the contrast agents.^{15,16} An overview of various ultrasound contrast agents and the corresponding data acquisition and post-processing techniques used in various preclinical and clinical applications is summarized in Figure 1.

Thus, this review article presents the various types of ultrasound contrast agents used in ultrasound molecular cancer imaging, highlighting advanced imaging techniques, offering an overview of ultrasound contrast agents, and introducing applications in current preclinical and clinical studies. Furthermore, this paper also discusses the challenges and future perspective of ultrasound molecular cancer imaging.

■ ULTRASOUND CONTRAST AGENTS

The main ultrasound contrast agents are microbubbles, nanobubbles, gas vesicles, and nanodroplets.^{17–19} Different ligands (including antibodies, peptides, carbohydrates, etc.) can be coupled to enhance their specificity. Ultrasound contrast agents can be targeted to intravascular or extravascular biomarkers depending on their sizes. To functionalize the ultrasound contrast agent, the targeting ligand can be integrated in the shell composition of the agent. Because of the difference in the contrast signals between ultrasound contrast agents and tissue, contrast agents can be clearly observed. Therefore, an increase in the contrast can be achieved. The comparison between various ultrasound contrast

agents used in ultrasound molecular imaging is summarized in Table 1.

Microbubbles. The most clinically used contrast agents in CEUS are small gas-filled spheres, which are usually encapsulated by a lipid shell layer. Microbubbles have a diameter ranging from 1 to 10 μm . Thus, they cannot extravasate outside the vasculature. At present, the main commercialized microbubble contrast agents include SonoVue (Bracco, Milano, Italy), DEFINITY (Lantheus Medical Imaging, Massachusetts, U.S.A.), and Optison and Sonazoid (GE Healthcare, Wisconsin, U.S.A.).²⁰ These contrast agents are all clinically approved by the Food and Drug Administration (FDA).

Nanobubbles. With the recent rapid developments in nanomedicine, nanobubbles, which are about 10 times smaller than microbubbles, can be manufactured as a new type of contrast agent.²¹ Nanobubbles are nanoparticles that are commonly formed by a lipid shell and a perfluorocarbon gas core.^{13,22,23} Nanobubbles can be classified as surface and bulk nanobubbles according to their different sizes. Surface nanobubbles are about tens of nanometers in height and about 1 μm in width. For bulk nanobubbles, the supersaturation of dissolved gas causes the generation of nanobubbles that are about 100 nm in size.¹⁹ Nanobubbles normally have lower contrast signals compared to microbubbles, but they are widely applied in therapeutic applications for various malignant tumors.^{23–25} The nano scale of nanobubbles makes them have the potential to be extravasated into the tumor space because of the enhanced permeability and retention (EPR) effect. The extravasation of these nanoscale contrast agents facilitates highly sensitive observation of tumor biomarkers, which is essential in early cancer diagnosis.^{26,27}

Gas Vesicles. Gas vesicles are biomolecular contrast agents for ultrasound imaging.²⁸ They are normally air-filled gas cores encapsulated by protein shells.^{29,30} Previous studies have demonstrated that gas vesicles have a better performance in cellular biocompatibility compared to the conventional microbubble contrast agents. They can last about 15 min under the continuous acoustic exposure. Additionally, the nanoscale gas vesicles enable them to extravasate into the tumor space because of EPR effects. Gas vesicles have been widely studied in the past and now have moved into the field of ultrasound imaging diagnosis and therapeutic research.²⁹

Nanodroplets. Nanodroplets can be activated by acoustic waves to transition from liquid droplets to gaseous bubbles. They are one of the most highly researched alternatives to clinically used microbubble contrast agents.³¹ The difference between nanobubbles and nanodroplets is the core composition. The core composition of a nanobubble is normally a low-boiling-point perfluorocarbon gas (e.g., C₃F₈ and C₄F₁₀), whereas the core composition of a nanodroplet is commonly a high-boiling-point perfluorocarbon liquid (e.g., C₅F₁₂ and C₆F₁₄) so that these nanodroplet contrast agents can be stable at physiological temperature. Conventional microbubbles cannot extravasate into a cancerous lesion via the gap junction between the endothelial cells.³² The advantages of applying nanodroplet contrast agents consist of smaller sizes, longer circulation time in the physiological environment, and the capability to be activated to provide ultrasound contrast signals compared to the most clinically used microbubble contrast agents. Furthermore, since their diameters are significantly smaller than those of microbubbles, nanodroplets can potentially extravasate into the cancerous tissue. Nanodroplets can be formed by the condensation of microbubbles. Figure 2 demonstrates the interaction between MDA-MB-231 breast cancer cells with targeted/nontargeted nanodroplets before and after ultrasonic activation. It can be clearly seen that, for the targeted nanodroplet, a few relatively large microbubbles were formed after ultrasonic activation.

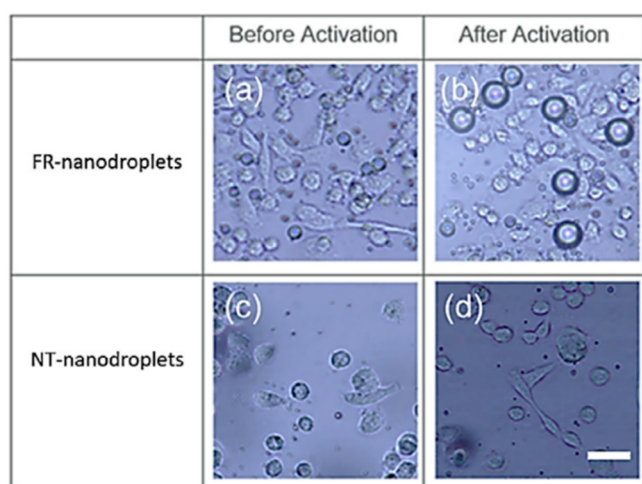


Figure 2. Microscopic imaging of folate receptor targeted nanodroplets and nontargeted nanodroplets with breast cancer cell lines before and after ultrasonic activation in vitro: (a) targeted nanodroplets before activation; (b) targeted nanodroplets after activation; (c) nontargeted nanodroplets before activation; (d) nontargeted nanodroplets after activation. The scale bar represents 30 μm.

ADVANCED DETECTION TECHNIQUES

The imaging contrast can be increased after intravenous injection of ultrasound contrast agents into the blood pool. The detection of the contrast signal relies on the nonlinear response of the contrast agents. After emitting acoustic waves from the ultrasound transducer, the ultrasound contrast agents will be hit and backscattered with a nonlinear signal, whereas tissues are more likely to send back linear signals. Therefore, the linear part of the signal from the tissue background is suppressed so as to increase the nonlinear signals from the ultrasound contrast agents injected into the bloodstream.³³ An overview of imaging techniques based on the nonlinearity of ultrasound contrast agents is shown in Figure 3.

Pulse Inversion. Pulse inversion consists of two imaging pulses. After transmitting the first imaging pulse, the later pulse is shifted by 180° in phase. After the reception of the two imaging pulses from the transducer, the responses of these two imaging pulses are summed up. Because of the nonlinearity, the summed response for tissue is close to zero, whereas the summed response for the ultrasound contrast agents does not cancel out.³⁴ Two signals are received and summed to suppress the linear signals; then, the difference between the nonlinear signals and background tissue signals can be clearly observed.

Amplitude Modulation. For the amplitude modulation method, two imaging pulses are transmitted. After transmitting the first imaging pulse, the second imaging pulse is transmitted with a half-amplitude compared to that of the first imaging pulse. These two pulses are then accumulated, as the later imaging pulse is multiplied by 2. The summed signals from the tissue are suppressed as the summed response from the contrast agents is not zero because of the nonlinearity.³⁵

Contrast Pulse Sequencing. Contrast pulse sequencing is an integration of the pulse inversion and amplitude modulation techniques.³⁶ After transmitting the first pulse, the later pulse is shifted 180° and only has a half-amplitude compared to the first pulse.

Singular Value Decomposition. Compared to the imaging techniques which improve the contrast signal during the data acquisition, singular value decomposition is an image postprocessing technique that can separate contrast and background signals based on the spatial and temporal properties of the ultrasound pulses.^{37,38} Briefly, the spatial and temporal properties of the acquired image data are combined by putting them into a Casorati matrix. All image frames are vectorized and reshaped as the corresponding columns. The detailed process can be seen in Figure 4. This Casorati matrix can be factorized by singular value decomposition (SVD) into the equation below:

$$\mathbf{S} = \mathbf{U}\mathbf{D}\mathbf{V}^*$$

where the columns of $\mathbf{U}$ are the spatial singular vectors and the columns of $\mathbf{V}$ are the corresponding temporal vectors. $\mathbf{D}$ is a diagonal matrix that contains elements which are the singular values. For the detection of contrast signals, a relatively lower singular value threshold is chosen to filter out the unwanted tissue and noise signals.³⁹

PRECLINICAL AND CLINICAL APPLICATIONS

Ultrasound molecular imaging has shown its great potential in the early detection of tumors, imaging monitoring during treatments, and drug delivery and therapy. To date, there are a number of preclinical and clinical applications being explored

Imaging Technique	Transmission		Reception		
	1 st Pulse	2 nd Pulse	1 st Pulse	2 nd Pulse	Final Response
Pulse Inversion			Tissue		---
			Contrast Agents		---
Amplitude Modulation			Tissue		---
			Contrast Agents		---
Contrast Pulse Sequencing			Tissue		---
			Contrast Agents		---

Figure 3. Overview of imaging techniques based on the nonlinearity of ultrasound contrast agents.

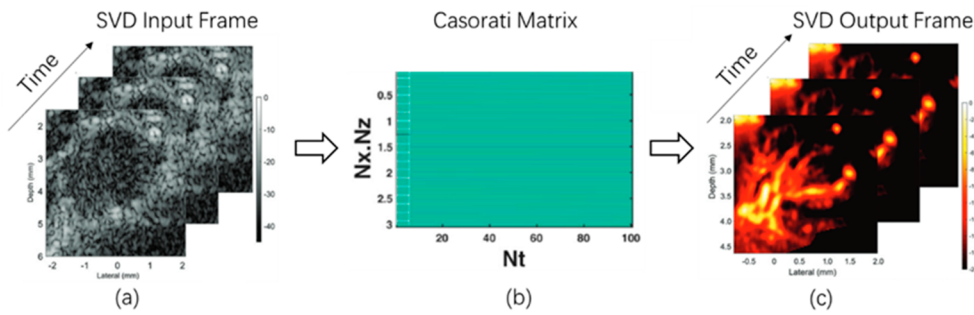


Figure 4. Generation of the Casorati matrix for singular value decomposition. Each image of spatial properties is reshaped as a column vector of the Casorati matrix. (a) The SVD input frame is a normal B-mode ultrasound image acquired from the ultrasound transducer. (b) The input frame is reshaped into a Casorati matrix. (c) The SVD output frame highlights the contrast signals within the vasculature.

Table 2. Ultrasound Molecular Imaging Detection of Cancers in Animal Models

type	binding ligands	diameter (nm)	application
nanobubble	prostate-specific membrane antigen(PSMA)	277 ± 11	detection of prostate cancer using male athymic Balb/c nude mice (ref 41)
nanobubble	carbonic anhydrase IX(CAIX)	478 ± 68	early diagnosis of various malignant tumors using nude mice (ref 23)
nanobubble	prostate-specific membrane antigen	274 ± 8	detection of prostate cancer using male athymic nude mice (ref 41)
nanobubble	SRC homology-2(SHP2)	535 ± 14	detection of thyroid cancer using nude mice (ref 61)
nanobubble	nucleolin (NCL)	459 ± 37	detection of triple-negative breast cancer using female BALB/c-nu mice (ref 62)
gas vesicle	poly(ethylene glycol)(PEG)	400–500	detection of Lewis lung carcinoma using SHO mice (ref 42)
gas vesicle	poly(ethylene glycol) (PEG) and hyaluronic acid (HA)		tumor detection using nude mice (ref 43)
microbubble	VEGFR2	519 ± 72	assessment of anti-VEGF antibody treatment using mice deficient in VEGFR1 (ref 45)
microbubble	VEGFR2		evaluation of early response to anti-VEGF therapy using C6 rat (ref 46)
nanodroplet	chitosan		imaging and treatment using male BALB/c mice (ref 47)
nanodroplet	organic anion transporting polypeptides(OATPs)		detection of cutaneous malignant melanoma using Oatp1a/Oatp1b null mice (ref 63)
microbubble	$\alpha\beta 3$ integrin	171 ± 57	detection of ovarian cancer using laying hens (ref 64)
microbubble	VEGFR2		detection of ovarian cancer using laying hens (ref 65)

to accelerate the clinical translation.^{8,40} Table 2 summarizes ultrasound molecular cancer imaging in various animal models.

Early Detection of Cancer. A previous study has demonstrated the first in-human ultrasound molecular imaging using BR55 contrast agents.⁸ A contrast enhancement of the prostate lesions could be clearly observed. BR55 is a vascular endothelial growth factor receptor 2 (VEGFR2)-targeted

ultrasound molecular contrast agent. Furthermore, the authors also showed that ultrasound molecular imaging using this contrast agent to assess the kinase insert domain receptor (KDR) on breast and ovarian cancers is clinically feasible and safe.⁴⁰ It has demonstrated meaningful results in various preclinical experiments regarding the ultrasound molecular imaging.

Besides using the clinical grade microbubble contrast agents, Wang et al. have shown that prostate-specific membrane antigen (PSMA)-targeted nanobubbles could increase the extravasation and retention level in PSMA-expressing tumors.⁴¹ Yan et al. have shown that, by coating gas vesicles with poly(ethylene glycol) (PEG) chains, the coated gas vesicles can improve recirculation and maintain the contrast enhancement during ultrasound molecular cancer imaging.⁴² Wang et al. integrated PEG and hyaluronic acid (HA) into the shells of gas vesicles and found that the coated gas vesicle was capable of escaping the filtering from the reticuloendothelial system (RES) and could penetrate cancerous vasculature via EPR effects.⁴³

Cancer Treatment Monitoring. Certain target molecules are overexpressed in tumors; therefore, the expression level of these target molecules can be evaluated by the quantification of the ultrasound contrast signals. Therefore, ultrasound molecular imaging can be applied to assess the results of antiangiogenic therapy, radiation therapy, and traditional chemotherapy.⁴⁴ This could be an effective approach for evaluating whether a certain treatment method is useful in patients with cancers. Korpanty et al. demonstrated that the contrast signals of anti-VEGFR2-targeted microbubbles decreased after treatment using anti-VEGF antibody.⁴⁵ Deshpande et al. showed the evaluation of the early assessment of anti-VEGF therapy in a mouse model using VEGFR2-targeted microbubble contrast agents.⁴⁶ It should be noted that the ultrasound molecular imaging signal changes before any morphological alternations. These studies have shown that ultrasound molecular cancer imaging may be an effective method for assessing early therapeutic response prior to any morphological changes. Therefore, a distinguishable clinical added value can be seen in assessing early treatment response by using targeted ultrasound molecular imaging.

Image-Guided Cancer Therapy. Ultrasound contrast agents not only play an important role in ultrasound molecular imaging, they are also considered as carriers to load drugs for relevant drug delivery and therapeutic applications. Zhou et al. have demonstrated that biocompatible chitosan nanodroplet (BCNDs) can be used to spatiotemporally control the delivery of doxorubicin (DOX) to promote the anticancer effects of DOX and inhibit the tumor development.⁴⁷ Lea-Banks et al. used pentobarbital-loaded nanodroplets for targeted neuro-modulation and localized delivery of pentobarbital.⁴⁸ Bar-Zion et al. used low-frequency ultrasound to convert gas vesicles into cavitating microbubbles so as to generate local mechanical effects on demand. In this case, gas vesicles not only acted as contrast agents for imaging but also played an important role as tissue-disrupting agents for cancer therapy.⁴⁹ The nature of ultrasound contrast agents makes them suitable as not only imaging agents but also drug carriers.

Image-Mediated Sonodynamic Therapy. Ultrasound-triggered sonodynamic therapy is a newly established therapeutic modality, since it can penetrate deeply in tissues.⁵⁰ Over the past decade, ultrasound molecular imaging-mediated sonodynamic therapy has been widely investigated.⁵¹ Yan et al. demonstrated that poly(lactic-co-glycolic acid) (PLGA) microcapsules integrated with the sonosensitizer hematoporphyrin monomethyl ether (HMME) nanoparticles could be used for not only contrast-enhanced ultrasound imaging but also a synergistic agent for sonodynamic therapy.⁵² Si et al. used phase-transformation nanoparticles to demonstrate that nanoparticle-mediated sonodynamic therapy can effectively enhance

the antitumor immune response of breast cancer.⁵³ With the increasing demand for precise and personalized therapeutic treatments, a large number of sonosensitizers with imaging capabilities have been investigated for significantly enhancing treatment efficacy.

Ultrasound Super-resolution Imaging. Inspired by optical super-resolution localization microscopy, ultrasound super-resolution imaging is an emerging ultrasound imaging technique which bypasses the compromise between penetration and resolution, as conventional ultrasound imaging is limited in image resolution by the diffraction to the wavelength scale. After the injection of ultrasound contrast agents, spatially isolated contrast signals can be localized, and their displacement can be tracked with subwavelength resolution. The ultrasound super-resolution imaging technique was first demonstrated *in vitro* using a single microchannel in 2011 to show that ultrasound super-resolution imaging could remarkably break the inherent diffraction limit.⁵⁴ This technique was then implemented in a mouse ear in 2015.⁵⁵ This study first introduced the concept of a super-resolution velocity map and showed that the magnitude and direction of flow velocity within the microvasculature can be derived. Moreover, ultrasound super-resolution imaging was demonstrated in a rat brain⁵⁶ and a cancer model⁵⁷ to show its potential to be a useful tool for a more comprehensive understanding of disease progressions that alter blood flow. In 2018, the first in-human applications of the ultrasound super-resolution imaging techniques using clinical scanners were demonstrated in human breast cancer.⁵⁸ Opacic et al. showed that ultrasound super-resolution imaging (US SRI) could provide a number of clinical indices that differentiate tumors with different microvascular characteristics, and it may provide opportunities for functional evaluation of tumors and the assessment of new therapeutic methods.

■ CHALLENGES AND FUTURE PERSPECTIVE

Ultrasound molecular imaging using contrast agents has shown great potential in disease diagnosis and characterization. However, there are a few challenges to be overcome to accelerate the clinical translation of ultrasound molecular imaging. In terms of the clinical translation of ultrasound molecular imaging and commercialization of ultrasound contrast agents, there also exist a few challenges. As a previous study reported, it normally costs \$350 million on average and about 6.5 years for academic research and *in vivo* animal experiments. Additionally, another 8 years on average and \$250 million are required to test an ultrasound contrast agent through clinical trials.⁵⁹

One of the challenges in the ultrasound molecular cancer imaging technique is imaging resolution. This is because the conventional ultrasound imaging resolution limit is about 10^{-3} m, whereas the image resolution required to observe chemical and molecular events is about 10^{-9} m.⁵⁹ Therefore, the development of advanced molecular imaging technique (e.g., molecular **ultrasound super-resolution imaging**) is expected to effectively improve the accuracy of disease diagnosis. By considering both high-sensitivity and high-resolution imaging, the lesion can be accurately described, and *in situ* imaging can be performed to guide the assessment. In terms of ultrasound contrast agents, for microbubble contrast agents, the size of these contrast agents makes them only suitable for intravascular imaging. For nanobubble contrast agents, since the size of a nanobubble is significantly smaller than that of a

microbubble, there will be a decrease in signal intensity. For gas vesicles, the preparation process for these contrast agents is relatively complex, which makes it relatively difficult to translate into clinics. Moreover, another barrier of applying gas vesicles in preclinical application is the rapid uptake of these gas vesicles into the RES and also the rapid decrease in contrast signals after infusion ends.⁴² **For nanodroplet contrast agents, the imaging of these contrast agents requires a further activation pulse to activate these contrast agents into imageable microbubble contrast agents.**

To date, the application of ultrasound molecular imaging has been demonstrated in a large number of preclinical studies and different animal models. Various contrast agents with different targeting ligands have been developed to understand the progression of different disease mechanisms. However, a better standardization of these targeted contrast agents is expected for acceleration of the clinical translation. This is because the development of a clinical grade ultrasound contrast agent requires large and extensive clinical trials to demonstrate safety issues. Therefore, the targeted microbubble contrast agent (e.g., BR55 microbubbles from SonoVue) is still preferred for use in clinical applications in ultrasound molecular imaging. For the other nanoscale contrast agents, they still require large-scale clinical trials to show applicability in humans to further accelerate the clinical translation. Furthermore, this academic research in newly found molecular targets faces a challenge in that industrial funding is normally offered for phase II/III clinical trials. This further makes it difficult to narrow down the gap of clinical translation.

SUMMARY

Driven by the rapid developments in molecular target design and advanced imaging methodologies, ultrasound molecular imaging has shown its great potential in a number of animal and preclinical studies to broaden the application of the conventional ultrasound imaging technique. The feasibility of applying these targeted ultrasound contrast agents in the early diagnosis of tumors, treatment monitoring, and image-guided therapy has been demonstrated. This review also discussed how these ultrasound contrast agents can be used in ultrasound super-resolution imaging, a newly emerging imaging technique, to significantly improve the image resolution of ultrasound images. These demonstrations showed the capability of these contrast agents to detect metabolic changes at the molecular level to improve visualization and quality of care. However, to be broadly accepted by clinicians, these ultrasound molecular imaging techniques will be required to offer a distinguishable added value compared to the current clinical CEUS techniques. To accelerate the clinical translation from academic research into current clinical applications, the translational process is required to be promoted by academia and pharmaceutical companies together. With the rapid developments in molecular chemistry, ultrasound physics, and imaging processing techniques, ultrasound molecular cancer imaging can be expected to evolve into a highly sensitive and specific imaging tool to help clinicians for better disease diagnosis and management.

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Author Contributions

G.Z. and H.-R.Y. contributed equally to this work. G.Z., conceptualization, literature review, writing—original draft; H.-R.Y., funding acquisition, resources, writing—original draft; Y.S., funding acquisition, project administration, writing—review and editing; Z.-Z.G., funding acquisition, project administration, resources, writing—review and editing.

Notes

The authors declare no competing financial interest.

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VOCABULARY

Ultrasound imaging, an imaging technique that uses an acoustic wave to view inside the body; microbubble contrast agents, gas-filled microbubbles administered intravenously to the systemic circulation to generate ultrasound contrast signals; contrast-enhanced ultrasound, an imaging technique that utilizes contrast agents to improve the visualization of blood vessels; molecular imaging, an imaging technique that focuses on imaging molecules of medical interest to obtain the corresponding molecular information; nanodroplet contrast agents, alternative ultrasound contrast agents used to generate ultrasound contrast signals post acoustic activation

REFERENCES

- (1) Abedi, M. H.; Yao, M. S.; Mittelstein, D. R.; Bar-Zion, A.; Swift, M. B.; Lee-Gosselin, A.; Barturen-Larrea, P.; Buss, M. T.; Shapiro, M. G. Ultrasound-controllable engineered bacteria for cancer immunotherapy. *Nat. Commun.* **2022**, *13* (1), 1585.
- (2) Gil, B.; Anastasova, S.; Yang, G. Z. Low-powered implantable devices activated by ultrasonic energy transfer for physiological monitoring in soft tissue via functionalized electrochemical electrodes. *Biosens. Bioelectron.* **2021**, *182*, 113175.
- (3) Sawyer, D. P.; Bar-Zion, A.; Farhadi, A.; Shivaie, S.; Ling, B.; Lee-Gosselin, A.; Shapiro, M. G. Ultrasensitive ultrasound imaging of

gene expression with signal unmixing. *Nat. Methods* **2021**, *18* (8), 945–952.

(4) Macé, E.; Montaldo, G.; Cohen, I.; Baulac, M.; Fink, M.; Tanter, M. Functional ultrasound imaging of the brain. *Nat. Methods* **2011**, *8* (8), 662–664.

(5) Kuznetsova, L. A.; Coakley, W. T. Applications of ultrasound streaming and radiation force in biosensors. *Biosens. Bioelectron.* **2007**, *22* (8), 1567–1577.

(6) Jiang, Q.; Zeng, Y.; Xu, Y.; Xiao, X.; Liu, H.; Zhou, B.; Kong, Y.; Saw, P. E.; Luo, B. Ultrasound Molecular Imaging as a Potential Non-invasive Diagnosis to Detect the Margin of Hepatocarcinoma via CSF-1R Targeting. *Frontiers in Bioengineering and Biotechnology* **2020**, *8*, 783.

(7) Abou-Elkacem, L.; Bachawal, S. V.; Willmann, J. K. Ultrasound molecular imaging: Moving toward clinical translation. *European journal of radiology* **2015**, *84* (9), 1685–1693.

(8) Smeenge, M.; Tranquart, F.; Mannaerts, C. K.; de Reijke, T. M.; van de Vijver, M. J.; Laguna, M. P.; Pochon, S.; de la Rosette, J.; Wijkstra, H. First-in-Human Ultrasound Molecular Imaging With a VEGFR2-Specific Ultrasound Molecular Contrast Agent (BR55) in Prostate Cancer: A Safety and Feasibility Pilot Study. *Invest Radiol* **2017**, *52* (7), 419–427.

(9) Köse, G.; Darguzyte, M.; Kiessling, F. Molecular Ultrasound Imaging. *Nanomaterials* **2020**, *10* (10), 1935.

(10) Luo, Y.; Fan, C.; Song, Y.; Xu, T.; Zhang, X. Ultra-trace enriching biosensing in nanoliter sample. *Biosens. Bioelectron.* **2022**, *210*, 114297.

(11) Ling, W.; Nie, J.; Zhang, D.; Yang, Q.; Jin, H.; Ou, X.; Ma, X.; Luo, Y. Role of Contrast-Enhanced Ultrasound (CEUS) in the Diagnosis of Cervical Lymph Node Metastasis in Nasopharyngeal Carcinoma (NPC) Patients. *Front. Oncol.* **2020**, *10*, 972.

(12) Challenging paradigms in tumour drug delivery. *Nat. Mater.* **2020**, *19* (5), 477–477.

(13) Zeng, F.; Du, M.; Chen, Z. Nanosized Contrast Agents in Ultrasound Molecular Imaging. *Front. Bioeng. Biotechnol.* **2021**, *9*, 758054.

(14) Punjabi, M.; Xu, L.; Ochoa-Espinosa, A.; Kosareva, A.; Wolff, T.; Murtaja, A.; Broisat, A.; Devoogdt, N.; Kaufmann, B. A. Ultrasound Molecular Imaging of Atherosclerosis With Nanobodies. *Arterioscler., Thromb., Vasc. Biol.* **2019**, *39* (12), 2520–2530.

(15) Anvari, A.; Forsberg, F.; Samir, A. E. A Primer on the Physical Principles of Tissue Harmonic Imaging. *RadioGraphics* **2015**, *35* (7), 1955–1964.

(16) Joseph, S.; Gronewold, T. M. A.; Schlensog, M. D.; Olbrich, C.; Quandt, E.; Famulok, M.; Schirner, M. Specific targeting of ultrasound contrast agent (USCA) for diagnostic application: an in vitro feasibility study based on SAW biosensor. *Biosens. Bioelectron.* **2005**, *20* (9), 1829–1835.

(17) Loskutova, K.; Grishenkov, D.; Ghorbani, M. Review on Acoustic Droplet Vaporization in Ultrasound Diagnostics and Therapeutics. *BioMed Res. Int.* **2019**, *2019*, 9480193.

(18) Schnell, C. Gas vesicles enable ultrasound imaging. *Nat. Methods* **2018**, *15* (3), 159–159.

(19) Bu, X.; Alheshibri, M. The effect of ultrasound on bulk and surface nanobubbles: A review of the current status. *Ultrasonics Sonochemistry* **2021**, *76*, 105629.

(20) Wu, S.-K.; Chu, P.-C.; Chai, W.-Y.; Kang, S.-T.; Tsai, C.-H.; Fan, C.-H.; Yeh, C.-K.; Liu, H.-L. Characterization of Different Microbubbles in Assisting Focused Ultrasound-Induced Blood-Brain Barrier Opening. *Sci. Rep.* **2017**, *7* (1), 46689.

(21) Perera, R. H.; Hernandez, C.; Zhou, H.; Kota, P.; Burke, A.; Exner, A. A. Ultrasound imaging beyond the vasculature with new generation contrast agents. *WIREs Nanomedicine and Nanobiotechnology* **2015**, *7* (4), 593–608.

(22) Li, Y.; Chen, X.; Zhou, Z.; Li, Q.; Westover, K. D.; Wang, M.; Liu, J.; Zhang, S.; Zhang, J.; Xu, B.; Wei, X. Dynamic surveillance of tamoxifen-resistance in ER-positive breast cancer by CAIX-targeted ultrasound imaging. *Cancer Med.* **2020**, *9* (7), 2414–2426.

(23) Zhu, L.; Wang, L.; Liu, Y.; Xu, D.; Fang, K.; Guo, Y. CAIX aptamer-functionalized targeted nanobubbles for ultrasound molecular imaging of various tumors. *Int. J. Nanomedicine* **2018**, *13*, 6481–6495.

(24) Yan, F.; Song, Z.; Du, M.; Klivanov, A. L. Ultrasound molecular imaging for differentiation of benign and malignant tumors in patients. *Quant. Imaging Med. Surg.* **2018**, *8* (11), 1078–1083.

(25) Yang, H.; Zhou, T.; Cai, W.; Yi, X.; Liu, X.; Wang, Y.; Zhang, L.; Duan, Y. Novel dual-mode nanobubbles as potential targeted contrast agents for female tumors exploration. *Tumour Biol.* **2016**, *37* (10), 14153–14163.

(26) Krupka, T. M.; Solorio, L.; Wilson, R. E.; Wu, H.; Azar, N.; Exner, A. A. Formulation and Characterization of Echogenic Lipid–Pluronic Nanobubbles. *Mol. Pharmaceutics* **2010**, *7* (1), 49–59.

(27) Wu, H.; Roguin, N. G.; Krupka, T. M.; Solorio, L.; Yoshiara, H.; Guenette, G.; Sanders, C.; Kamiyama, N.; Exner, A. A. Acoustic Characterization and Pharmacokinetic Analyses of New Nanobubble Ultrasound Contrast Agents. *Ultrasound in Medicine & Biology* **2013**, *39* (11), 2137–2146.

(28) Farhadi, A.; Ho, G. H.; Sawyer, D. P.; Bourdeau, R. W.; Shapiro, M. G. Ultrasound imaging of gene expression in mammalian cells. *Science* **2019**, *365* (6460), 1469–1475.

(29) Ling, B.; Lee, J.; Maresca, D.; Lee-Gosselin, A.; Malounda, D.; Swift, M. B.; Shapiro, M. G. Biomolecular Ultrasound Imaging of Phagolysosomal Function. *ACS Nano* **2020**, *14* (9), 12210–12221.

(30) Long, H.; Qin, X.; Xu, R.; Mei, C.; Xiong, Z.; Deng, X.; Huang, K.; Liang, H. Non-Modified Ultrasound-Responsive Gas Vesicles from Microcystis with Targeted Tumor Accumulation. *Int. J. Nanomedicine* **2021**, *16*, 8405–8416.

(31) Sheeran, P. S.; Matsuura, N.; Borden, M. A.; Williams, R.; Matsunaga, T. O.; Burns, P. N.; Dayton, P. A. Methods of Generating Submicrometer Phase-Shift Perfluorocarbon Droplets for Applications in Medical Ultrasonography. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* **2017**, *64* (1), 252–263.

(32) Zhang, G.; Harput, S.; Hu, H.; Christensen-Jeffries, K.; Zhu, J.; Brown, J.; Leow, C. H.; Eckersley, R. J.; Dunsby, C.; Tang, M. X. Fast Acoustic Wave Sparsely Activated Localization Microscopy: Ultrasound Super-Resolution Using Plane-Wave Activation of Nanodroplets. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* **2019**, *66* (6), 1039–1046.

(33) Chen, H.; Evangelou, D.; Loskutova, K.; Ghorbani, M.; Grishenkov, D. On the Development of a Novel Contrast Pulse Sequence for Polymer-Shelled Microbubbles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* **2021**, *68* (5), 1569–1579.

(34) Khairalseed, M.; Oezdemir, I.; Hoyt, K. Contrast-enhanced ultrasound imaging using pulse inversion spectral deconvolution. *J. Acoust. Soc. Am.* **2019**, *146* (4), 2466–2474.

(35) Rabut, C.; Wu, D.; Ling, B.; Jin, Z.; Malounda, D.; Shapiro, M. G. Ultrafast amplitude modulation for molecular and hemodynamic ultrasound imaging. *Appl. Phys. Lett.* **2021**, *118* (24), 244102.

(36) Brown, K. G.; Hoyt, K. Evaluation of Nonlinear Contrast Pulse Sequencing for Use in Super-Resolution Ultrasound Imaging. *IEEE Trans Ultrason Ferroelectr Freq Control* **2021**, *68* (11), 3347–3361.

(37) Brown, J.; Christensen-Jeffries, K.; Harput, S.; Zhang, G.; Zhu, J.; Dunsby, C.; Tang, M. X.; Eckersley, R. J. Investigation of Microbubble Detection Methods for Super-Resolution Imaging of Microvasculature. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* **2019**, *66* (4), 676–691.

(38) Lok, U. W.; Song, P.; Trzasko, J. D.; Daigle, R.; Borisch, E. A.; Huang, C.; Gong, P.; Tang, S.; Ling, W.; Chen, S. Real time SVD-based clutter filtering using randomized singular value decomposition and spatial downsampling for micro-vessel imaging on a Verasonics ultrasound system. *Ultrasonics* **2020**, *107*, 106163.

(39) Hasegawa, H.; Nagaoka, R.; Omura, M.; Mozumi, M.; Saito, K. Investigation of feasibility of singular value decomposition clutter filter in plane wave imaging with packet transmission sequence. *J. Med. Ultrason (2001)* **2021**, *48* (1), 13–20.

- (40) Willmann, J. K.; Bonomo, L.; Testa, A. C.; Rinaldi, P.; Rindi, G.; Valluru, K. S.; Petrone, G.; Martini, M.; Lutz, A. M.; Gambhir, S. S. Ultrasound Molecular Imaging With BR55 in Patients With Breast and Ovarian Lesions: First-in-Human Results. *J. Clin. Oncol.* **2017**, *35* (19), 2133–2140.
- (41) Wang, Y.; De Leon, A. C.; Perera, R.; Abenojar, E.; Gopalakrishnan, R.; Basilion, J. P.; Wang, X.; Exner, A. A. Molecular imaging of orthotopic prostate cancer with nanobubble ultrasound contrast agents targeted to PSMA. *Sci. Rep.* **2021**, *11* (1), 4726.
- (42) Yan, J.; Yin, M.; Foster, F. S.; Démore, C.E. M. Tumor Contrast Imaging with Gas Vesicles by Circumventing the Reticuloendothelial System. *Ultrasound in Medicine and Biology* **2020**, *46* (2), 359–368.
- (43) Wang, G.; Song, L.; Hou, X.; Kala, S.; Wong, K. F.; Tang, L.; Dai, Y.; Sun, L. Surface-modified GVs as nanosized contrast agents for molecular ultrasound imaging of tumor. *Biomaterials* **2020**, *236*, 119803.
- (44) Kaneko, O. F.; Willmann, J. K. Ultrasound for molecular imaging and therapy in cancer. *Quant. Imaging Med. Surg.* **2012**, *2* (2), 87–97.
- (45) Korpanty, G.; Sullivan, L. A.; Smyth, E.; Carney, D. N.; Brekken, R. A. Molecular and Clinical Aspects of Targeting the VEGF Pathway in Tumors. *J. Oncol.* **2010**, *2010*, 652320.
- (46) Deshpande, N.; Pysz, M. A.; Willmann, J. K. Molecular ultrasound assessment of tumor angiogenesis. *Angiogenesis* **2010**, *13* (2), 175–188.
- (47) Zhou, X.; Guo, L.; Shi, D.; Meng, D.; Sun, X.; Shang, M.; Liu, X.; Zhao, Y.; Li, J. Ultrasound-responsive highly biocompatible nanodroplets loaded with doxorubicin for tumor imaging and treatment in vivo. *Drug Delivery* **2020**, *27* (1), 469–481.
- (48) Lea-Banks, H.; O'Reilly, M. A.; Hamani, C.; Hynynen, K. Localized anesthesia of a specific brain region using ultrasound-responsive barbiturate nanodroplets. *Theranostics* **2020**, *10* (6), 2849–2858.
- (49) Bar-Zion, A.; Nourmahnad, A.; Mittelstein, D. R.; Shivaie, S.; Yoo, S.; Buss, M. T.; Hurt, R. C.; Malounda, D.; Abedi, M. H.; Lee-Gosselin, A.; Swift, M. B.; Maresca, D.; Shapiro, M. G. Acoustically triggered mechanotherapy using genetically encoded gas vesicles. *Nat. Nanotechnol.* **2021**, *16* (12), 1403–1412.
- (50) Guo, J.; Pan, X.; Wang, C.; Liu, H. Molecular Imaging-Guided Sonodynamic Therapy. *Bioconjugate Chem.* **2022**, *33*, 993.
- (51) Paefgen, V.; Doleschel, D.; Kiessling, F. Evolution of contrast agents for ultrasound imaging and ultrasound-mediated drug delivery. *Frontiers in Pharmacology* **2015**, *6*, 197.
- (52) Yan, S.; Lu, M.; Ding, X.; Chen, F.; He, X.; Xu, C.; Zhou, H.; Wang, Q.; Hao, L.; Zou, J. HematoPorphyrin Monomethyl Ether polymer contrast agent for ultrasound/photoacoustic dual-modality imaging-guided synergistic high intensity focused ultrasound (HIFU) therapy. *Sci. Rep.* **2016**, *6* (1), 31833.
- (53) Si, Y.; Yue, J.; Liu, Z.; Li, M.; Du, F.; Wang, X.; Dai, Z.; Hu, N.; Ju, J.; Gao, S.; Wang, X.; Yuan, P. Phase-Transformation Nanoparticle-Mediated Sonodynamic Therapy: An Effective Modality to Enhance Anti-Tumor Immune Response by Inducing Immunogenic Cell Death in Breast Cancer. *International journal of nanomedicine* **2021**, *16*, 1913–1926.
- (54) Couture, O.; Besson, B.; Montaldo, G.; Fink, M.; Tanter, M. Microbubble ultrasound super-localization imaging (MUSLI). *2011 IEEE International Ultrasonics Symposium* **2011**, 1285–1287.
- (55) Christensen-Jeffries, K.; Browning, R. J.; Tang, M. X.; Dunsby, C.; Eckersley, R. J. In vivo acoustic super-resolution and super-resolved velocity mapping using microbubbles. *IEEE Trans. Med. Imaging* **2015**, *34* (2), 433–40.
- (56) Errico, C.; Pierre, J.; Pezet, S.; Desailly, Y.; Lenkei, Z.; Couture, O.; Tanter, M. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* **2015**, *527* (7579), 499–502.
- (57) Lin, F.; Shelton, S. E.; Espíndola, D.; Rojas, J. D.; Pinton, G.; Dayton, P. A. 3-D Ultrasound Localization Microscopy for Identifying Microvascular Morphology Features of Tumor Angiogenesis at a Resolution Beyond the Diffraction Limit of Conventional Ultrasound. *Theranostics* **2017**, *7* (1), 196–204.
- (58) Opacic, T.; Dencks, S.; Theek, B.; Piepenbrock, M.; Ackermann, D.; Rix, A.; Lammers, T.; Stickeler, E.; Delorme, S.; Schmitz, G.; et al. Motion model ultrasound localization microscopy for preclinical and clinical multiparametric tumor characterization. *Nat. Commun.* **2018**, *9* (1), 1527.
- (59) Liu, C. H.; Sastre, A.; Conroy, R.; Seto, B.; Pettigrew, R. I. NIH Workshop on Clinical Translation of Molecular Imaging Probes and Technology—Meeting Report. *Molecular Imaging and Biology* **2014**, *16* (5), 595–604.
- (60) Yin, T.; Wang, P.; Zheng, R.; Zheng, B.; Cheng, D.; Zhang, X.; Shuai, X. Nanobubbles for enhanced ultrasound imaging of tumors. *Int. J. Nanomedicine* **2012**, *7*, 895–904.
- (61) Hu, Z.; Yang, B.; Li, T.; Li, J. Thyroid Cancer Detection by Ultrasound Molecular Imaging with SHP2-Targeted Perfluorocarbon Nanoparticles. *Contrast Media Mol. Imaging* **2018**, 8710862.
- (62) Jugniot, N.; Massoud, T. F.; Dahl, J. J.; Paulmurugan, R. Biomimetic nanobubbles for triple-negative breast cancer targeted ultrasound molecular imaging. *J. Nanobiotechnol.* **2022**, *20*, 267.
- (63) Yang, H.; Cai, W.; Lv, W.; Zhao, P.; Shen, Y.; Zhang, L.; Ma, B.; Yuan, L.; Duan, Y.; Yao, K. A new strategy for accurate targeted diagnosis and treatment of cutaneous malignant melanoma: dual-mode phase-change lipid nanodroplets as ultrasound contrast agents. *Int. J. Nanomedicine* **2019**, *14*, 7079–7093.
- (64) Barua, A.; Yellapa, A.; Bahr, J. M.; Machado, S. A.; Bitterman, P.; Basu, S.; Sharma, S.; Abramowicz, J. S. VEGFR2-Targeted Ultrasound Imaging Agent Enhances the Detection of Ovarian Tumors at Early Stage in Laying Hens, a Preclinical Model of Spontaneous Ovarian Cancer. *Ultrason. Imaging* **2015**, *37* (3), 224–237.
- (65) Barua, A.; Yellapa, A.; Bahr, J. M.; Adur, M. K.; Utterback, C. W.; Bitterman, P.; Basu, S.; Sharma, S.; Abramowicz, J. S. Interleukin 16- (IL-16-) Targeted Ultrasound Imaging Agent Improves Detection of Ovarian Tumors in Laying Hens, a Preclinical Model of Spontaneous Ovarian Cancer. *Biomed. Res. Int.* **2015**, 567459.